The Impossibility of Faithful Explanation Under Underdetermination

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Abstract

When a system is underspecified—when multiple configurations produce the same observable output—any explanation faces an inherent tradeoff. We prove that no explanation can simultaneously be faithful (consistent with the system's native explanation), stable (invariant across observationally equivalent configurations), and decisive (committing to every distinction the native explanation makes). This impossibility holds whenever the Rashomon property obtains: equivalent configurations with incompatible explanations. We prove the Rashomon property is both necessary and sufficient—within our framework, the exact boundary between possibility and impossibility. The theorem is instantiated across nine domains: feature attribution, attention maps, counterfactual explanations, concept probes, causal discovery, model selection, saliency maps, language model explanations, and mechanistic interpretability. For causal discovery, the Rashomon property is derived from first principles (Markov equivalence). The resolution—G-invariant aggregation—is proved Pareto-optimal among stable methods. The result is mechanically verified in the Lean 4 proof assistant (346 theorems, 0 unproved goals).

Significance. Explainability is central to trustworthy AI, scientific inference, and regulatory compliance. We prove a universal impossibility: no method of explaining an underspecified system can simultaneously be faithful to the system, stable across equivalent configurations, and decisive in its commitments. This formalizes a core consequence of the Quine–Duhem underdetermination thesis as a mathematical theorem, to our knowledge for the first time. The impossibility applies to nine major explanation paradigms in machine learning—from SHAP values to neural network circuit analysis—and extends to scientific explanation, medical diagnosis, and legal interpretation. The constructive resolution (aggregate over equivalent configurations) provides a Pareto-optimal tradeoff among stable explanation methods. The entire proof is mechanically verified, eliminating logical gaps in the formal proofs.

Keywords: impossibility theorem, explainability, underdetermination, Rashomon effect, formal verification

1 Introduction

When multiple configurations of a system produce identical observable output, any attempt to explain the system's behavior faces a fundamental constraint. This paper proves that constraint is an impossibility: no explanation can simultaneously be faithful, stable, and decisive.

This structure arises across science. In philosophy of science, Quine Quine [1951] and Duhem Duhem [1954] argued that empirical evidence underdetermines theory choice—no finite body of observations can uniquely select a scientific theory from among empirically adequate alternatives. In neuroscience, Marder and Goaillard Marder and Goaillard [2006] documented that nervous systems with identical functional output are built from qualitatively different circuit configurations, a phenomenon called *neural degeneracy* Edelman and Gally [2001]. In statistics and machine learning, Breiman's Rashomon effect Breiman [2001] and its modern elaboration by Rudin et al. Rudin et al. [2024] describe the multiplicity of near-optimal models with divergent internal structure. In economics, Manski Manski [2003] developed partial identification theory precisely because observational data generically fail to pin down a unique model. In database theory, Bancilhon and Spyratos Bancilhon and Spyratos [1981] proved that view updates are inherently ambiguous when the view is not injective.

We prove the impossibility, characterize its exact boundary (the Rashomon property—necessary and sufficient), demonstrate it across nine domains with the most detailed treatment in machine learning, and provide a constructive resolution. Informally:

> *If two configurations of a system produce the same observable output but honestly report incompatible explanations, then no explanation method can simultaneously be faithful to the system, stable across equivalent configurations, and decisive in its commitments.*

The proof is four steps—decisiveness, stability, faithfulness, contradiction—and is mechanically verified in the Lean 4

proof assistant de Moura and Ullrich [2021] (73 files, 346 theorems, 72 axioms, 0 `sorry`). The core impossibility requires zero axioms beyond the Rashomon property itself.

Our contributions are: (1) To our knowledge, this is a universal impossibility theorem unifying nine domain-specific results under a single proof; (2) a proof that the Rashomon property is both necessary and sufficient—the exact boundary between possibility and impossibility; (3) nine instances spanning feature attribution, attention maps, counterfactual explanations, concept probes, causal discovery (with Rashomon *derived* from Markov equivalence), model selection, saliency maps, language model explanations, and mechanistic interpretability; (4) a constructive resolution via G-invariant aggregation proved uniquely optimal; (5) empirical validation across seven instances; and (6) complete mechanical verification. The result connects to Hunt–Stein invariant decision theory Lehmann and Romano [2005] and, to our knowledge, provides the first mathematical formalization of the Quine–Duhem underdetermination thesis.

2 Framework

2.1 Explanation Systems

We introduce a minimal abstract framework in which the impossibility can be stated and proved. All nine instances in Section 3 are special cases.

Definition 1 (Explanation System)**.** *An* explanation system *is a tuple* $\mathcal{S} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{observe}, \mathsf{explain}, \perp)$ *where:*

- $\mathbf{\Theta}$ *is a* configuration space*—the set of possible configurations of the system (e.g., model parameters, experimental setups, circuit wirings);*
- $\mathcal{H}$ *is an* explanation space*—the set of possible explanations;*
- Y *is an* observable space*—the set of measurable outputs;*
- $\mathsf{observe} : \mathbf{\Theta} \to Y$ *assigns to each configuration its observable behavior;*
- $\mathsf{explain} : \mathbf{\Theta} \to \mathcal{H}$ *assigns to each configuration the explanation it "honestly reports"; and*
- $\perp \subseteq \mathcal{H} \times \mathcal{H}$ *is an* incompatibility relation *satisfying* irreflexivity*:* $\neg \perp (h, h)$ *for every* $h \in \mathcal{H}$*.*

An *explanation map* $E : \mathbf{\Theta} \to \mathcal{H}$ is a procedure that a practitioner *chooses*. The native explanation $\mathsf{explain}(\theta)$ records what the configuration actually implies; $E(\theta)$ is what the chosen method reports.

2.2 Three Desiderata

Definition 2 (Faithful, Stable, Decisive)**.** *An explanation map* E *is:*

- ***Faithful*** *if* $E(\theta)$ *never contradicts the native explanation:* $\forall\, \theta,\ \neg \perp (E(\theta), \mathsf{explain}(\theta))$*.*
- ***Stable*** *if equivalent configurations receive the same explanation:* $\forall\, \theta_1, \theta_2,\ \mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2) \Rightarrow E(\theta_1) = E(\theta_2)$*.*
- ***Decisive*** *if* $E(\theta)$ *inherits every incompatibility of the native explanation:* $\forall\, \theta, h,\ \perp (\mathsf{explain}(\theta), h) \Rightarrow \perp (E(\theta), h)$*.*

Faithfulness says the method does not contradict the system. Stability says the method is reproducible—it does not depend on which equivalent configuration happens to be examined. Decisiveness says the method is at least as committal as the system itself: whatever the native explanation rules out, the chosen explanation also rules out.

2.3 The Rashomon Property

Definition 3 (Rashomon Property)**.** *An explanation system has the* Rashomon property *if there exist* $\theta_1, \theta_2 \in \mathbf{\Theta}$ *with*

$$\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2) \quad \textit{and} \quad \perp (\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2)).$$

Two configurations produce the same observable output yet honestly report incompatible explanations. This is the formal statement of underdetermination at the explanation level.

2.4 Main Theorem

Theorem 4 (Universal Explanation Impossibility)**.** *If an explanation system has the Rashomon property, then no explanation map* E *can simultaneously be faithful, stable, and decisive.*

Proof. Let θ_1, θ_2 witness the Rashomon property. Suppose E is faithful, stable, and decisive.

1. **Decisiveness** at θ_1: $\perp (\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2))$ implies $\perp (E(\theta_1), \mathsf{explain}(\theta_2))$.
2. **Stability**: $\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2)$ implies $E(\theta_1) = E(\theta_2)$, so $\perp (E(\theta_2), \mathsf{explain}(\theta_2))$.
3. **Faithfulness** at θ_2: $\neg \perp (E(\theta_2), \mathsf{explain}(\theta_2))$.
4. Steps 2 and 3 contradict. □

Proposition 5 (Tightness)**.** *Each pair of properties is simultaneously achievable: (i) faithful + decisive (set* $E = \mathsf{explain}$*; stability fails); (ii) faithful + stable (set* E *to a neutral constant; decisiveness fails); (iii) stable + decisive (set* E *to a maximally committal constant; faithfulness fails).*

Proposition 6 (Necessity)**.** *An explanation system admits a faithful, stable, and decisive explanation map if and only if it does* not *have the Rashomon property.*

The Rashomon property is therefore the *exact boundary* between possibility and impossibility. The proofs of Propositions 5–6 are in *SI Appendix*.

3 Instances

We instantiate Theorem 4 in four domains chosen for cross-disciplinary breadth. Five additional instances (attention maps, counterfactual explanations, concept probes, model selection, language model explanations) are detailed in *SI Appendix*.

3.1 Feature Attribution (Machine Learning)

Fix $P \geq 2$ features with collinearity groups. The configuration space $\mathbf{\Theta}$ is the set of trained models (e.g., gradient-boosted trees, Lasso, neural networks); the observable space Y is the input–output function; the explanation space $\mathcal{H}$ is the set of attribution vectors $(\varphi_1, \ldots, \varphi_P)$ assigning importance to each feature; and the incompatibility relation declares two vectors incompatible when they rank the same collinear pair in opposite orders. Collinearity induces a Rashomon set of near-optimal models that differ only in which feature within a correlated group acts as the primary predictor D'Amour et al. [2022], Fisher et al. [2019]. The impossibility follows from Theorem 4: no attribution method (SHAP, LIME, Integrated Gradients) can simultaneously be faithful, stable, and decisive. This subsumes the result of Bilodeau et al. [2024] as a special case.

For gradient-boosted trees with per-group correlation ρ, the attribution ratio diverges as $1/(1-\rho^2)$ as $\rho \to 1$. For Lasso at any $\rho > 0$, one correlated feature is zeroed out entirely (infinite ratio). Empirically, SHAP rankings flip at a 33% rate across equivalent models on standard benchmarks.

3.2 Causal Discovery

The causal discovery instance is the strongest because the Rashomon property is *derived from first principles* rather than axiomatized.

The configuration space $\mathbf{\Theta}$ is the set of directed acyclic graphs (DAGs) over a fixed vertex set V. The observable space Y is the set of conditional independence (CI) structures—each element records, for every triple $(X_i, X_j \mid S)$, whether $X_i \perp\!\!\!\perp X_j \mid S$ holds. The observation map sends each DAG to the CI structure it encodes via d-separation (the Markov condition). Two DAGs map to the same CI structure if and only if they are *Markov equivalent* Verma and Pearl [1991]. The explanation map is any orientation rule that returns a fully directed graph. The incompatibility relation declares two DAGs incompatible when they orient at least one edge in opposite directions.

Proposition 7 (Causal Rashomon Property—Derived)**.** *For any completed partially directed acyclic graph (CPDAG) containing at least one undirected edge $\{X, Y\}$, there exist Markov-equivalent DAGs h, h' with opposite orientations of that edge. Hence the causal explanation system has the Rashomon property.*

Proof. An undirected edge in the CPDAG means both orientations are Markov-consistent. Let h orient it as $X \to Y$ and h' as $Y \to X$. Then $\mathsf{observe}(h) = \mathsf{observe}(h')$ (same CI structure) and $\perp(h, h')$ (opposite orientations). □

The three-variable chain $X \to Y \to Z$ and fork $X \leftarrow Y \to Z$ provide the canonical example: both encode $X \perp\!\!\!\perp Z \mid Y$ (and no other CI relation), so they are Markov equivalent, yet they disagree on the X–Y edge direction. No causal discovery algorithm can distinguish them from observational data alone.

By Theorem 4, no orientation rule can simultaneously be faithful (return the true DAG), stable (give the same answer for Markov-equivalent structures), and decisive (always commit to a fully oriented graph). This is the classical Markov equivalence underdetermination recast in our framework, with the Rashomon property derived rather than assumed. The resolution is to report CPDAGs Chickering [2002] rather than DAGs—trading decisiveness for faithfulness and stability, exactly as the theorem predicts.

3.3 Mechanistic Interpretability

Mechanistic interpretability (MI) aims to reverse-engineer neural networks into human-understandable circuits Elhage et al. [2022], Conmy et al. [2023], Anthropic [2025]. The configuration space is the set of weight configurations; the observable space is the input–output function; the explanation space is the set of circuit decompositions; and the incompatibility relation declares two circuits incompatible when they attribute the same computation to contradictory subsets of network components.

Meloux et al. [2025] found **85 distinct valid circuits** with zero circuit error for a single XOR task, each admitting an average of 535.8 valid interpretations. Complementary evidence from Bricken et al. [2025] shows that sparse autoencoders trained on the same data learn different features depending on the random seed.

By Theorem 4, no circuit explanation method can simultaneously be faithful (reflect the actual internal circuitry), stable (assign the same circuit to functionally equivalent networks), and decisive (commit to a single circuit). This result does not undermine MI—it clarifies that the goal should be to characterize the *space* of valid circuits (the "circuit equivalence class") rather than to search for a single canonical decomposition. This mirrors the CPDAG resolution for causal discovery: report which circuit features are invariant across all valid decompositions (the "skeleton") and which are decomposition-dependent (the "reversible edges").

For AI safety, the implication is direct: any safety argument that depends on a specific circuit being THE mechanism of a capability (e.g., deception) must account for the possibility that functionally equivalent alternative circuits exist, potentially with different safety-relevant properties.

Table 1: The three escape routes from the explanation trilemma.

Drop	Lose	Gain	Tools
Faithful	May contradict	Stable + decisive	Fixed rules
Stable	Changes on retrain	Faithful + decisive	SHAP, single circuit
Decisive	Ties, uncertainty	Faithful + stable	DASH, CPDAG

3.4 Neural Degeneracy (Neuroscience)

Marder and Goaillard Marder and Goaillard [2006] documented that crustacean stomatogastric ganglia producing identical motor patterns are built from neurons with qualitatively different ionic conductance profiles. Edelman and Gally Edelman and Gally [2001] formalized this as *degeneracy*: structurally distinct elements performing the same function. In our framework, the configuration space is the set of neural circuit parameterizations (ionic conductances, synaptic weights); the observable space is the motor pattern; and the explanation space assigns to each circuit a mechanistic account of which ion channels drive the rhythm.

Degeneracy is precisely the Rashomon property: two circuit configurations produce the same rhythmic output yet honestly report incompatible mechanistic explanations (e.g., "the rhythm is driven by the I_{Ca} current" versus "the rhythm is driven by the I_{h} current"). By Theorem 4, no mechanistic account of neural circuit function can simultaneously be faithful to the specific circuit, stable across degenerate circuits, and decisive about which ionic mechanism is primary. This connects directly to the MI instance (Section 3.3): artificial and biological neural circuits face the same structural impossibility.

The remaining five instances—attention maps (60% argmax flip rate across equivalent DistilBERT models), counterfactual explanations (23.5% direction flip rate), concept probes (0.90 direction instability), model selection (80% best-model flip rate), and language model explanations (34.5% citation flip rate)—are detailed with full experimental methodology in *SI Appendix*.

4 Resolution

4.1 Escape Routes

The impossibility is tight: one desideratum must be sacrificed. Table 1 summarizes the three escape routes.

4.2 G-Invariant Aggregation

Let G be a group acting on the configuration space such that $g \cdot \theta$ preserves observables: $\mathsf{observe}(g \cdot \theta) = \mathsf{observe}(\theta)$. An explanation map is *G-invariant* if $E(g \cdot \theta) = E(\theta)$ for all g, θ.

Proposition 8 (G-Invariance Implies Stability)**.** *If G acts transitively on observe-fibers and E is G-invariant, then E is stable.*

When G is finite and the explanation space admits averaging, the canonical G-invariant map is the *orbit average*:

$$\bar{E}(\theta) = \frac{1}{|G|} \sum_{g \in G} \mathsf{explain}(g \cdot \theta).$$

This map is stable (by G-invariance), faithful in expectation (the expected explanation matches the expected native explanation over the orbit), and Pareto-optimal among stable maps. It sacrifices decisiveness: where the orbit disagrees, the average produces ties.

The connection to classical statistics is direct. The Hunt–Stein theorem Lehmann and Romano [2005] establishes that among all invariant decision procedures, the best invariant procedure is admissible. Our G-invariant resolution is the explanation-theoretic instantiation: among all stable explanation maps, the orbit average achieves the best possible faithfulness.

4.3 Instance-Specific Resolutions

The abstract resolution instantiates concretely across domains:

- **Feature attribution**: DASH (**D**iversified **A**ggregation of **SH**AP)—average SHAP values over an ensemble of retrained models. Consensus equity ($\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$ for symmetric features) is proved in Lean 4.
- **Causal discovery**: report the CPDAG (completed partially directed acyclic graph) rather than a single DAG Chickering [2002]. Undirected edges mark the unresolvable queries.
- **Mechanistic interpretability**: report the circuit equivalence class—the set of all valid circuits—rather than a single decomposition. Invariant circuit features form the "skeleton."
- **Neural degeneracy**: characterize the *degenerate set* of ionic conductance profiles producing the target motor pattern, rather than committing to a single mechanistic account.
- **Medical diagnosis**: differential diagnosis *is* G-invariant resolution—reporting the set of conditions consistent with the symptoms rather than committing to a single diagnosis.

5 Empirical Validation

We validate the impossibility empirically across seven of the nine instances (Table 2). For each instance, we train multiple models that are functionally equivalent (matched on prediction accuracy) and measure explanation instability.

Table 2: Cross-instance empirical validation. All instability measures have 95% bootstrap CIs excluding zero. Prediction equivalence confirms functional identity.

Instance	Metric	Value	Pred. equiv.
Attribution	SHAP flip	33%	>99%
Attention	Argmax flip	60%	100%
Counterfactual	Dir. flip	23.5%	ΔAUC<0.03
Concept probe	$1-\lvert\cos\rvert$	0.90	97.2%
Model selection	Best flip	80%	ΔAUC<0.02
Saliency map	Peak flip	9.6%	78.8%
LLM explanation	Citation flip	34.5%	100%

Three results deserve emphasis. First, the 60% attention argmax flip rate: despite 100% prediction agreement among 10 DistilBERT models on SST-2, the most-attended token changes for the majority of sentences. This is not noise—it is the Rashomon property in action. Second, the 0.90 concept direction instability: concept activation vectors extracted from 15 equivalent MLP classifiers are nearly orthogonal (mean $|\cos| = 0.10$), meaning the "concept direction" is almost entirely a function of the random seed, not the function the model computes. Third, the 0% instability for negative controls (instances where the Rashomon property does not hold, e.g., unique-solution linear regression), confirming that the theorem's boundary condition is empirically sharp.

Full experimental details—datasets, model architectures, perturbation protocols, confidence intervals, and negative controls—are in *SI Appendix*.

6 Discussion

6.1 Philosophy of Science: Formalizing Quine–Duhem

The Quine–Duhem thesis Quine [1951], Duhem [1954] holds that no scientific hypothesis can be tested in isolation: empirical evidence always underdetermines theory choice because auxiliary assumptions can be adjusted to accommodate any observation. Van Fraassen van Fraassen [1980] elevated this to a central argument for constructive empiricism. Despite a century of philosophical discussion, the thesis has never been formalized as a mathematical theorem with a precise boundary condition.

Theorem 4 provides that formalization. The configuration space $\mathbf{\Theta}$ corresponds to the space of theories; the observable space Y to empirical data; the explanation map to the interpretive framework; and the Rashomon property to the existence of empirically equivalent theories with incompatible explanations. The theorem states: underdetermination at the observable level propagates to underdetermination at the explanation level, and no interpretive procedure can escape the structural tradeoff. The necessity result (Proposition 6) sharpens the thesis: the impossibility holds *if and only if* the Rashomon property obtains, providing the exact boundary that philosophical formulations lack.

Okasha Okasha [2011] connected Kuhn's theory choice to Arrow's impossibility theorem Arrow [1951] in social choice. Our result provides a complementary connection: underdetermination of explanation is an impossibility theorem in the same structural sense as Arrow's, with the Rashomon property playing the role of Arrow's independence of irrelevant alternatives.

6.2 AI Regulation

The EU AI Act EU Parliament [2024] requires "meaningful explanations" for high-risk systems. SR 11-7 Office of the Comptroller of the Currency [2011] requires "developmental evidence" documenting model decisions. Fairness regulations face a parallel impossibility: Chouldechova Chouldechova [2017] proved that common fairness metrics cannot simultaneously be satisfied when base rates differ across groups. Both implicitly demand faithfulness, stability, and decisiveness simultaneously. Theorem 4 shows this is impossible whenever models are underspecified—which is the norm for modern machine learning systems. The resolution is not to abandon regulation but to reframe it: require disclosure of *which* desideratum is relaxed and why, mandate G-invariant explanations (e.g., ensemble-averaged attributions) with explicit uncertainty quantification, and accept that ties where the data are ambiguous are a feature of honest explanation, not a deficiency.

6.3 AI Safety

The mechanistic interpretability instance (Section 3.3) has direct implications for AI safety. If a safety argument depends on a specific circuit being the mechanism of a dangerous capability, the impossibility theorem requires that the argument account for alternative circuits—functionally equivalent decompositions that may have different safety-relevant properties. The 85 valid circuits found by Meloux et al. [2025] for a simple XOR task suggest that real-world safety-critical capabilities will admit far more. The constructive response is to build safety arguments on circuit-invariant properties—features that hold across the entire equivalence class—rather than on any single decomposition.

6.4 Closing

The impossibility does not counsel despair but precision: state which property is sacrificed, aggregate optimally, and report uncertainty. The Rashomon property is the exact boundary. When it holds, the trilemma is inescapable. When it does not, all three properties are jointly achievable. The task for the practitioner is to diagnose which regime obtains and to communicate the tradeoff honestly.

Materials and Methods

The Lean 4 formalization (73 files, 346 theorems, 72 axioms, 0 `sorry`) is available at [repository URL]. Experimental code and data are in *SI Appendix*. The core impossibility theorem (`explanation_impossibility`) depends on zero axioms beyond the Rashomon property. Full axiom inventory and theorem list are in *SI Appendix*, Table S1.

Acknowledgments

[To be added.]

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